

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25

**Visual Working Memory in Early Development:
A Developmental Cognitive Neuroscience Perspective**

Aaron T. Buss^a, Shannon Ross-Sheehy^a, & Greg D. Reynolds^{a,*}

Author Affiliations:
^a University of Tennessee
Department of Psychology
Knoxville, TN. 37996

***Corresponding Author:** Greg D. Reynolds: greynolds@utk.edu

26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51

Abstract

In this paper, we review the literature on the development of visual working memory (VWM). We focus on two major periods of development, infancy and early childhood. First, we discuss the innovative methods that have been devised to understand how the development of selective attention and perception provide the foundation of VWM abilities. We detail the behavioral and neural data associated with the development of VWM during infancy. Next, we discuss various signatures of development in VWM during early childhood in the context of spatial and featural memory processes. Here we focus on the developmental transition to more adult-like VWM properties. Finally, we discuss computational frameworks that have explained the complex patterns of behavior observed in VWM tasks from infancy to adulthood and attempt to explain links between measures of infant VWM and childhood VWM.

Visual Working Memory in Early Development:**A Developmental Cognitive Neuroscience Perspective**

Visual short-term memory (VSTM) is a capacity limited memory system involved in the brief storage of visual information. Because VSTM is fundamentally involved in other processes, such as perception and learning, it has been recognized as a visual working memory (VWM) system (Luck, 2008; Oakes, Baumgartner, Barrett, Messenger, & Luck, 2013). VWM allows us to actively represent our visual environment for the purpose of mental tasks such as planning, comparison, and response selection (Baddeley, 1986), and is functionally and anatomically distinct from long-term memory (Baddeley & Hitch, 1974). Though we know much about the development of long-term memory over the first few years of life (for reviews, see: Bauer, 2009; Cowan, 1995; Nelson, 1995; Pelphrey & Reznick, 2003; Reynolds, 2015; Rose, Feldman, & Jankowski, 2004; Rovee-Collier & Cuevas, 2009), we are only just beginning to understand the development of VWM in infancy and beyond. In this paper, we review research on VWM in infancy and early childhood. We begin by reviewing foundational research on early visual behavior and highlighting the role of lower-level visual processes in VWM. We then briefly review tasks exploring VWM development in infants and children, discussing important task-based considerations. Finally, we discuss neurally grounded computational models of VWM, and highlight their important contributions to our understanding of mechanisms of developmental change. Throughout each section, we focus on research examining both behavioral and neural correlates of VWM.

Origins of research on infant visual memory

Though infant VSTM is a relatively young research area, it comes from a long tradition of research examining simple visual behaviors in preverbal infants. This pioneering work by Robert Fantz and others was foundational, and established critical parameters for using simple visual preferences to infer perceptual and even cognitive development (Colombo & Fagen, 1990; Fantz, 1964). In a now famous series of studies, infants were presented with a pair of images, one on the left and one on the right, and duration of looking to each image was measured. On subsequent trials, one of the images remained the same, while the other was swapped with a new image (locations of the novel and familiar images varied randomly from trial to trial). As memory for the familiar image grew, looking to the novel stimulus increased. Thus, the appearance of a *novelty preference* could be taken as evidence for memory of the familiar image. This was some of the earliest work to systematically explore the development of visual memory in infancy.

Fantz theorized that these kinds of early visual exploratory behaviors may be necessary for typical postnatal development. If this is the case, he reasoned that infant eye movements must be *selective*, peripheral and central visual structures must be sufficiently developed to support *discrimination*, and infants must be able to *remember* what they have seen (Fantz, 1964). As the demonstration of even a moderate novelty preference requires attainment of all three competencies, memory is inextricably linked to *selectivity* and *discrimination*. Following is a brief review on the development of each of these visual competencies, and how they influence the emergence of VSTM.

99 *Visual Selectivity.* For visual exploration to fuel early learning and development, eye movements
100 must be *selective*—that is, they must reliably be directed to events and objects that are currently
101 relevant, such as a parent’s face, or a bottle being moved into view. Some of this selectivity is
102 apparent in the first weeks of life, and can be accomplished through simple bottom-up processing
103 of perceptual characteristics like motion, luminance contrast, or salience (Johnson, 1992;
104 Johnson, Posner, & Rothbart, 1991; Nagata & Dannemiller, 1996). Though the exact
105 mechanisms of attentional selectivity in infancy are not as well-understood (Johnson, Posner, &
106 Rothbart, 1991; Reynolds, Courage, & Richards, 2013; Reynolds & Roth, 2018), these early
107 biases are typically thought to be largely *involuntary*.

108
109 Development from 3 to 6 months is characterized by significant gains in the *voluntary* control of
110 eye movements (Johnson, 1990), and this selectivity is often described as a property of
111 endogenous, or goal-directed visual attention (Bahrick & Lickliter, 2014; Yantis & Jonides,
112 1990). Though specific brain areas may support aspects of attention such as switching or
113 vigilance (Frick et al., 1999; Posner & Petersen, 1990), attentional selectivity in general is most
114 likely a distributed property of the entire brain (Desimone & Duncan, 1995; Reynolds, 2015).
115 The rapid improvement in eye movement control across this age range is likely tied to further
116 development of a network of structures involved in orienting, including: the pulvinar nucleus of
117 the thalamus, posterior parietal areas, and the frontal eye-fields (Johnson et al., 1991; Posner &
118 Rothbart, 2013; Reynolds & Romano, 2016). These improvements in visual orienting and
119 selective attention foster more efficient perceptual processing and memory (e.g., Reynolds &
120 Richards, 2005, 2017; Richards, 1997), which in turn further facilitates learning.

Attentional selectivity may also be critical for VSTM, particularly when infants are young. For example, previous research examining the development of memory in infants has revealed that although VSTM improves rapidly between the 6th and 10th month of life (Oakes, Ross-Sheehy, & Luck, 2006; Ross-Sheehy, Oakes, & Luck, 2003), it is highly constrained before 7.5 months of age (Oakes et al., 2006). However, when an attention cue is used to direct attention, even 5-month-old infants demonstrate memory for stimulus arrays that are beyond their memory abilities when their attention is not cued (Ross-Sheehy, Oakes, & Luck, 2011). Infants also appear to be able to update the contents of VSTM, a process which necessarily requires selective attention (Moher & Feigenson, 2013; Yamaguchi & Feigenson, 2008). This clearly demonstrates a critical interaction between selective attention mechanisms and VSTM, and this dynamic interaction influences looking behavior as early as 5 months of age.

Visual Discrimination. The widespread adoption of infant visual preference paradigms resulted in a veritable boon of research examining the perceptual abilities of infants. This body of work suggests that most key visual perceptual abilities develop very early in infancy. For example, by 4 weeks of age, infants can reliably discriminate colors such as red and green, even when controlling for brightness (Hamer, Alexander, & Teller, 1982), and by three months, infants are likely trichromatic (i.e., fully functioning short, medium, and long wavelength cones), and can discriminate red, green and yellow (Adams & Courage, 1995). Infant acuity similarly undergoes rapid postnatal development. By three months, techniques that rely on reflexive orienting behaviors (optokinetic nystagmus or OKN, and visual evoked potentials, or VEP) suggest Snellen equivalencies that range from 20/150 to around 20/300 (Dobson & Teller, 1978). Although 3-month-old infants have surprisingly well-developed color vision and acuity (at least

for relatively close distances), these impressive abilities are partially mediated by relatively low contrast sensitivity; a typical 3-month-old requires approximately 50 times more luminance contrast than an adult to detect fine details in an image (Banks & Salapatek, 1981; Brown & Lindsey, 2009). However, given most functionally significant objects can be identified on the bases of their high contrast low spatial frequency information (e.g., the face of a conspecific), even early infant learning can be relatively robust.

Visual memory. It is likely that the early visual behaviors described above are influenced by limitations in VSTM. When an infant looks at an object, she or he begins to form a memory representation (Roder, Bushnell, & Sasseville, 2000). This rapidly formed memory likely resides in VSTM, where capacity limits may influence the fidelity of the representation. When the infant then subsequently looks to a new image, this fading memory trace may interact with the current focus of gaze. Comparison via a VSTM mechanism such as this may support the detection of novelty, further influencing looking behavior. Though this interplay between previously fixated and currently fixated items is clearly memory dependent, deficits in either attention or discrimination may also influence both the fidelity of the memory trace, and the subsequent detection of familiarity or novelty. Thus, the appearance of deficits in memory may be driven by limits in VSTM (e.g., capacity limits, decay, interference), by limits in attentional selectivity, or by limits in perceptual discrimination. Here we focus on the development of VSTM in infancy and beyond, in an attempt to identify key neural and behavioral developmental events.

Memory in infancy

Short-term memory or working memory? As stated previously, work with adults suggests VWM is a rapidly decaying short-term “working” memory store, marked by its highly limited capacity of around three to four integrated objects (Baddeley & Hitch, 1994; Baddeley & Logie, 1999). Though the term “working memory” is often used with tasks that include some sort of behavioral response (i.e., remember the location, sort the cards, press a button, etc.), the terms “working memory” and “visual short-term memory” are often conflated in both the infant and adult literatures. In the adult literature, this distinction is largely theoretical, as most adult tasks require adults to *act* on their memory (e.g., press a button if you noticed a color change). To the extent that working memory is a short-term memory used in service of a task (e.g., a decision followed by a button press), then nearly all adult tasks test both working memory and short-term memory. However, there is growing consensus that working memory may also operate *below* the level of awareness (Hassin, Bargh, Engell, & McCulloch, 2009; Joyce, 2016). This is important for work exploring infant VSTM, and may help bridge the gap between infant, toddler, and adult tasks.

The term “working memory” as it is now commonly understood was first used by Atkinson and Shiffrin to describe the function of their hypothesized “short-term store” (R. C. Atkinson & Shiffrin, 1968). This short-term store was distinct from both the long-term store and the sensory register, and had some unique properties including rapid decay, and limited capacity. This model was important in that it supported the distinction of long- and short-term memory stores, an idea that had fallen out of favor at the time. Drawing on the strengths of the Atkinson and Shiffrin model, Baddeley and Hitch (1974) re-conceptualized working memory as a multimodal system consisting of two short-term memory stores (the “phonological loop” for auditory information

and the “visuospatial sketchpad” for visual information) under the attentional control of a “central executive.” This inclusion of an attentional control process subsequently broadened the role of working memory to include critical “on line” cognitive processes such as reasoning, problem solving, response selection, and decision making (note, this model was later elaborated to include a multi-modal “episodic buffer”; see Baddeley, 2007). Though the Baddeley and Hitch (1974) model remains prominent, there are now multiple alternative frameworks, each of which has generated much empirical support (e.g., Cowan, 2005; Kane & Engle, 2002; Klingberg et al., 2000).

Importantly, one common feature of all of these models, is the inclusion of some aspect of attentional or “executive” control. In adult tasks, this executive control is typically measured using some sort of behavioral response, however it is not clear that the behavioral response or explicit awareness of the task is required, *per se*. Recent studies utilizing “one-shot” change detection tasks indicate that infants can utilize information stored in VSTM in the service of other cognitive processes (Mitsven, Cantrell, Luck, & Oakes, 2018; Oakes et al., 2013), which according to some views would qualify VSTM as a working memory system (e.g., Baddeley, 1986; Luck & Vogel, 2013). For example, Mitsven and colleagues (2018) designed a cued visual search task to test the possibility that 10-month-old infants’ VSTM representations influence their subsequent looking behavior. Infants were shown a single 500 ms presentation of a central stimulus followed by a 300 ms delay and then a test array of two items presented to either side of the central fixation point. One item in the test array matched the “one-shot” stimulus presumably stored in VSTM and the other item did not. Across two experiments utilizing colored squares and complex objects as stimuli, 10-month-olds were found to be more likely to look at the non-

matching item than the matching item. These findings indicate that the infants' VSTM guided their subsequent visual search during the test trials. Thus, it is likely that VSTM tasks do tap into VWM processes in infancy (Mitsven et al., 2018; also see, Kibbe, 2015). These findings also indicate that VSTM can direct infant selective attention and support the idea that attention and memory are linked in a bidirectional manner by mid- to late-infancy.

Improvements in VWM across infancy: Capacity and features

Capacity limits constrain the amount of information that one can keep active in memory, which has important consequences for cognitive processes that involve comparison across multiple exemplars (e.g., learning to recognize a parent's face, learning to discriminate a bottle from toys, etc.). Despite its obvious importance, there has been relatively little work addressing this topic in infancy, partly because it is difficult to isolate VSTM from other longer-term memory mechanisms. One key issue stems from the use of a novelty preference as test criterion, and the relatively long encoding durations typically required for this behavioral response.

In an attempt to address this constraint, researchers have begun to rely on rapid, serial presentations, during which arrays of colored squares appear and disappear rapidly (e.g., 500ms on, 250ms off, 500ms on, etc.). Using a modified paired-comparison procedure, infants are presented with two blinking arrays, one on the left and one on the right. In one array, the color of a single, randomly selected square changes at every onset. On the other array, the colors of the squares remain constant across each onset (see Figure 1 top panel). If infants can rapidly encode all the squares in the array and remember that information across the 250ms delay interval, then they should notice when one changes color, and look preferentially to that side. In other words,

they should show a *change preference*. Results across a series of studies using this approach have revealed VSTM develops rapidly the first year of life. As shown in Figure 1, 6.5-month-old infants are able to detect change for only set size 1 arrays, whereas older infants can detect change for arrays up to 3 or 4 items (Oakes, Messenger, Ross-Sheehy, & Luck, 2009; Oakes et al., 2006; Ross-Sheehy et al., 2003). Additional studies have revealed that these capacity limits in young infants are not driven by perceptual limitations (Ross-Sheehy et al., 2003), and performance is not improved even when every square in the changing arrays changes at every onset (Oakes et al., 2006). These dramatic deficits appear to be at least partially driven by a failure to bind color and location (Oakes et al., 2009), which may be ameliorated by the addition of an attention cue (Ross-Sheehy et al., 2011). More recent work with eye tracking suggests that like adults, infant memory is about the same for simple features as it is for integrated objects (Kwon, Luck, & Oakes, 2014), and encoding speed rivals that of adult participants (Oakes et al., 2013). Although some similarities exist between certain characteristics of VSTM for infants and adults, at the end of the infancy period VSTM is still relatively immature.

Insert Figure 1 here

Improvements in VWM across infancy: Duration and stability. Though change preference tasks typically are used to assess VSTM capacity for simple features such as color and are typically assessed using computerized tasks, other aspects of memory improve rapidly between 6 and 9 months and can more readily be tested using explicit behavioral responses such as reaching. For example, in the A-not-B task, infants are presented with two or more hiding wells. While the infant is watching, an object is hidden inside one of these wells. After a brief delay, the infant is allowed to reach to retrieve the object. After a series of successful retrievals from a given well,

the hiding place is switched to a different well. The A-not-B error refers to the strong tendency of young infants to continue to reach to the original well (Location A) even after watching the experimenter switch hiding places to the second well (Location B). Infants are less likely to show this perseverative error after 8 - 9 months of age (Cuevas & Bell, 2011; Hofstadter & Reznick, 1996; Pelphrey et al., 2004).

Diamond (1985; 1990) has proposed the higher success rate for older infants on the A-not-B task is based on gains in inhibitory control tied to maturation of dorsolateral prefrontal cortex (dlPFC). Support for the possibility that younger infants may lack sufficient inhibitory motor control and cognitive flexibility for successful A-not-B performance has come from studies demonstrating that when infants' gaze behavior and looking behavior do not match on a given trial, they are more likely to gaze at the correct hiding location than to reach for the correct hiding location (Hofstadter & Reznick, 1996). Importantly, Smith, Thelen, Titzer, and McLin (1999) showed that beyond a lack of inhibitory control, several contextual factors contribute to the A-not-B error for younger infants, including the number of training trials, the number of reaches to the A location prior to the switch, and the participant's posture on a given trial. Thus, although the A-not-B task clearly involves VWM, successful task performance also likely involves inhibitory processes and a certain level of cognitive flexibility.

In order to examine neural correlates of VWM and inhibitory control in infancy, Bell and colleagues have utilized the electroencephalogram (EEG) while infants perform looking versions of the A-not-B task (e.g., Bell, 2012; Bell & Adams, 1999; Bell & Wolfe, 2007; Cuevas & Bell, 2011). From these studies, it has been shown that both baseline frontal EEG power and task-

related changes in EEG power from 6 – 9 Hz are correlated with successful performance on the A-not-B task (Bell, 2002; Bell & Fox, 1994). Frontal-parietal and frontal-occipital EEG coherence values also correlate with successful A-not-B performance (Bell, 2012). In a recent study, Bacher, Retz, Lindon, and Bell (2017) utilized spontaneous blinking as an indirect measure of dopaminergic function and found that variability in spontaneous blink rate was positively associated with 10-month-olds' VWM performance on the A-not-B task.

At a broad level, these findings support the possibility that a frontal-parietal network and the dopaminergic neurotransmitter system are involved in VWM in infancy. This is consistent with findings from comparative research showing increased extracellular dopamine in prefrontal cortex for non-human primates during performance of a VWM task (Watanabe, Kodama, & Hikosaka, 1997). Research utilizing oculomotor delayed response tasks has found evidence of spatial tuning of neurons in dPFC associated with visuo-spatial WM in non-human primates (Goldman-Rakic, 1995; Rao, Williams, & Goldman-Rakic, 1999). Delay period activity within spatially selective cells in dPFC has been proposed to represent the neural basis of spatial visuo-spatial WM (Funahashi, Bruce, & Goldman-Rakic, 1989). However, according to Sreenivasan, Curtis, and D'Esposito (2014), sensory cortices maintain specific representations of VWM content while lPFC maintains representations of goal-related variables that may be involved in attentional control aspects of VWM.

Until recently, research examining functional neural activation in human participants during early development faced many practical challenges. However, recent applications using functional near-infrared spectroscopy (fNIRS) have allowed researchers to begin to observe

functional neural activation in infants and children. fNIRS overcomes many of the challenges of using fMRI to measure cortical blood-flow using a lightweight cap with embedded near-infrared laser emitters and detectors. By monitoring light intensity changes as near-infrared light is absorbed by oxy- and deoxy-hemoglobin, this technology can measure regional changes in blood oxygen concentration as infants and children perform a cognitive task. For example, Baird et al. (2002) measured activation from lateral frontal cortex of 10- to 12-month-olds in the A-not-B task and reported that infants who correctly searched at the B location on B trials showed stronger activation in lateral frontal cortex compared to infants who perseverated on reaching to the A location. Unfortunately, both fNIRS and EEG/source analysis techniques for neuroimaging are primarily constrained to activity produced in cortical regions and cannot provide evidence regarding the potential contributions of subcortical regions to a given process. However, findings from VWM studies utilizing EEG (e.g., Bell, 2012; Cuevas & Bell, 2011) and fNIRS (Baird et al., 2002) with infant participants are consistent with findings from comparative research (Goldman-Rakic, 1995; Watanabe et al., 1997) and neuroimaging research on older children and adults indicating areas of dlPFC, ventrolateral prefrontal cortex (vlPFC), intraparietal cortex, and posterior parietal cortex are recruited during VWM tasks (Courtney, Ungerleider, Keil, & Haxby, 1997; Crone, Wendelken, Donohue, van Leijenhorst, & Bunge, 2006; Fuster, 1997; Klingberg, Forssberg, & Westerberg, 2002; Scherf, Sweeney, & Luna, 2006; Sweeney et al., 1996).

Improvements in VWM: Binding features to locations. Binding in infants has been explored using a modified change preference task (Oakes et al., 2009, 2006). As in Ross-Sheehy et. al (2003), infants are shown two arrays, each consisting of three colored squares that blinked on and off for 20 seconds (500ms on, 300ms off, 500ms on, etc.). In one array, the colors of the three squares

stay the same at every onset, in the other array the color-location bindings for all three squares are swapped at every onset. If infants bind color to location in VSTM, they should notice when the bindings change, and look preferentially to the changing side. Note that this task should be trivially easy, as infants need only encode a single square to detect a change. Results demonstrate that 6.5-month-old infants show no evidence of recognizing the binding swap, whereas infants just one month older do (Oakes et al., 2006).

The ability of infants to bind objects to locations has also been examined in a series of studies by Kaldy and colleagues (Kaldy, Guillory, & Blaser, 2016; Kaldy & Leslie, 2003; Kaldy & Leslie, 2005; Kaldy & Sigala, 2004). For example, Kaldy and Leslie (2003, 2005) designed an occlusion task aimed at testing infants' ability to perform both identification and individuation processes involved in VWM. Infants are familiarized with two objects and then the objects are each placed behind an occluder. The occluders are then removed after a delay period. On change trials, removal of the occluders reveals the objects in reversed locations from their original hiding spots. On no-change trials, the objects remain in their original hiding locations. Infant looking behavior is measured in response to change trials versus no-change trials. Longer looking on change trials indicates successful individuation of the object which involves binding of object and location in VWM. Across studies, findings indicate that 6-month-olds are only able to bind object to location for the last object placed behind an occluder, whereas 9-month-olds are able to bind object to location for both objects (Kaldy & Leslie, 2003, 2005). Kaldy and Leslie (2005) proposed that gains in VWM performance on occlusion tasks that occur from 6 to 9 months of age are related to further development of medial temporal lobe structures, such as entorhinal cortex and parahippocampal cortex.

Research utilizing fNIRS to examine neural correlates of object processing in infancy has provided support for the possibility that regions of temporal cortex are involved in infant VWM for occluded objects (e.g., Wilcox & Biondi, 2015; Wilcox, Haslup, & Boas, 2010; Wilcox, Stubbs, Hirshkowitz, & Boas, 2012). For example, Wilcox and colleagues (2012) tested 3- to 5- and 11- to 12-month-old infants in an occlusion task in which objects moved behind a screen and then reappeared on the other side of the screen. Infants were tested in three conditions: shape difference, color difference, or control. In the shape difference condition, the object changed shape but not color upon reappearance. In the color difference, the object changed color but not shape. The object remained unchanged in the control condition.

Analysis of the fNIRS data revealed that both age groups demonstrated activation of anterior temporal areas in difference conditions but not in the control condition. However, 3- to 5-month-olds only demonstrated differential activation of anterior temporal cortex based on changes in shape, whereas 11- to 12-month-olds demonstrated differential activation based on both shape and color. Based on the anterior temporal findings, the authors concluded that between 5 and 11 months of age, infants develop the ability to individuate objects based on VWM for both shape and color features. Younger infants also demonstrated activation of parietal areas on shape difference trials, an effect that was not found in 11- to 12-month-olds in any of the three conditions. According to Wilcox and colleagues (2012), this activation of parietal areas was exclusive to the younger age group because at 3 to 5 months of age infants are dependent on motion-carried information to extract information regarding shape. By 11 to 12 months of age,

infants are able to extract, and retain in VWM, both shape and color features of objects without relying on motion-based information.

In summary, infancy is characterized by major gains in VWM. Figure 2 tracks developmental trends in performance on the behavioral and neural measures of VWM discussed above. Findings across multiple infant tasks clearly indicate significant improvement in VWM from approximately 4 to 12 months of age with the most notable improvements coming after 6 months of age (Cuevas & Bell, 2010; Diamond, 1990; Hofstadter & Reznick, 1996; Kaldy et al., 2016; Kaldy & Leslie, 2003; Kaldy & Leslie, 2005; O’Gilmore & Johnson, 1995; Oakes et al., 2009, 2006; Pelphrey et al., 2004; Reznick, Morrow, Goldman, & Snyder, 2004; Ross-Sheehy et al., 2003). The improvements in VWM that occur primarily across the second half of the first postnatal year include: an increase in VSTM capacity measured using change preference tasks from 1 item to 3 to 4 items (Oakes et al., 2006, 2009; Ross-Sheehy et al., 2003), increased ability to bind features (such as shape and color) and location of objects in occlusion and change preference tasks (Kaldy & Leslie, 2003, 2005; Oakes et al., 2006, 2009), and improved performance on the A-not-B task (Cuevas & Bell, 2010; Hofstadter & Reznick, 1996; Pelphrey et al., 2004).

Insert Figure 2 here

There has been some inconsistency in findings from studies measuring neural correlates of VWM in infancy. There is evidence from studies utilizing the A-not-B task that a frontal-parietal network and the dopaminergic neurotransmitter system are critically involved in successful performance (Bacher et al., 2017; Baird et al., 2002; Bell, 2012). In contrast, results from occlusion studies examining object individuation and VWM for features and locations highlight

the importance of anterior temporal areas for successful performance (Wilcox et al., 2010, 2012). The inconsistencies in findings across studies may be due to the possibility that these tasks tap into different aspects of VWM, and in contrast to change preference and object identification tasks, the A-not-B task likely involves an inhibitory or attentional control component that goes beyond basic VWM processes (Kaldy & Leslie, 2003). However, there is currently no clear consensus in the field on exactly what, if anything, separates VSTM from VWM in infancy.

Mitsven and colleagues (2018) have claimed that VSTM qualifies as a VWM system because it has been found to influence subsequent visual search behavior (also see Gilmore & Johnson, 1995). However, according to some, the ability to maintain task-relevant information in the face of task-irrelevant information is a key aspect of VWM that may not be met by basic visual search behavior (e.g., Vogel, McCollough, & Machizawa, 2005). Additionally, the length of familiarization on object identification tasks (e.g., Kaldy & Leslie, 2003, 2005) may push the limits of VSTM and instead tap into aspects of long-term memory. Further work focused on disentangling the neural mechanisms involved in VSTM, VWM, and long-term memory tasks in infancy is needed to help resolve issues related to whether these tasks tap into similar or distinct neural and cognitive processes. There has been a great deal of research on VWM in early childhood that provides insight into mechanisms involved in the development of VWM, which we now turn our focus to (for related research from middle childhood on, see Amso & Scerif, 2015; Astle & Scerif, 2011; Awh, Vogel, & Oh, 2006; Shimi, Nobre, Astle, & Scerif, 2014).

Working Memory in Early Childhood

Improvements in VWM: Capacity and features. Early childhood is a period during which children's performance on spatial WM and VWM tasks improves to equivalent levels as adults. Measures of VWM during this period are predictive of intelligence and academic performance (Raghubar, Barnes, & Hecht, 2010; Rose, Feldman, Jankowski, & Rossem, 2008) suggesting that these changes influence important aspects of general cognitive functioning. Although working memory develops into a distinct aspect of cognitive functioning in adulthood (Miyake et al., 2000), during early childhood measures of working memory are highly correlated with measures of inhibitory control (Tsujiimoto, Kuwajima, & Sawaguchi, 2007; Wiebe et al., 2011). Change detection tasks can be used to test capacity limits in toddlers. Unlike infants, however, toddlers can indicate whether a given trial is a "same" trial or a "different" trial. In the context of the change detection task, children's VWM capacity grows from around 1.5 objects at age 3 to a more adult-like 3-4 objects at age 7 (Riggs, McTaggart, Simpson, & Freeman, 2006; Simmering, 2012b, 2016; Simmering, Miller, & Bohache, 2015). Interestingly, performance across *same* and *different* trials improves in a systematic fashion during this period. Very young children tend to respond with "change" at a higher rate than older children and adults. This leads to some children showing more frequent false-alarm errors relative to miss errors. As a consequence, 3-year-olds perform more poorly on no-change trials but better on change trials (Buss, Fox, Boas, & Spencer, 2014; Simmering, 2016; Simmering et al., 2015). Moreover, performance on tasks involving color is better than tasks involving shapes, and performance with familiar colors is better than performance with less familiar colors (Simmering et al., 2015). Lastly, children's estimation of remembered color features becomes more precise (Simmering & Miller, 2016).

Improvements in spatial WM: Duration and stability. Unique signatures of development during early childhood have also been identified in the domain of spatial WM. Between the ages of 3 and 6, children's responses on spatial recall show systematic patterns of errors. In spatial recall tasks, a location is briefly cued by presenting a stimulus, then after a delay, participants are instructed to indicate where the cue was presented. This task has been implemented with a reference line to divide a homogenous space. Early in development, children's responses are biased toward the reference lines. By the age of 6, however, children's biases shift away from the midline similar to adults' spatial recall (Schutte & Spencer, 2009). Similar to feature memory recall, spatial recall also shows increases in precision during early childhood (Schutte, Spencer, & Schoner, 2003). Lastly, children's spatial recall is influenced by habits established in the context of a task similar to the A-not-B error. That is, Spencer and Schutte (2004) administered a task in which a toy was hidden within a homogeneous and continuous space to 2- to 4-year-olds. After observing a toy being hidden at an A location over a series of trials, the toy was then hidden at a B location and children were allowed to search for the object. On these B trials, children's searches were biased toward the A location where they observed the toy being hidden.

Improvements in VWM: Neuroimaging results. Neuroimaging studies with older children and adults have revealed a network of regions that are involved with different aspects of VWM, including encoding, maintenance, and comparison. These regions include, intraparietal sulcus (IPS), tempo-parietal junction (TPJ), and lateral prefrontal cortex (LPFC), among other regions (e.g., Crone et al., 2006; D'Esposito et al., 1999; Courtney et al., 1998; Fuster, 1997; Klingberg et al., 2002; Scherf et al., 2006; and Sweeney et al., 1996; Wijeakumar, Huppert, Magnotta, Buss, & Spencer, 2017). Buss, Fox, Boas, and Spencer (2014) recorded fNIRS data from

466 bilateral frontal and parietal cortices while 3- and 4-year-old children performed a change
467 detection task with shape stimuli. Children showed task-related activation in frontal and parietal
468 regions. Typically, a relationship between performance and hemodynamic activity is observed
469 with adult participants such that activation in parietal cortex increases as a function of set-size
470 and asymptotes once the number of items exceeds capacity. However, activation in parietal
471 cortex increased up to set-size 3 despite capacity estimates below 2 items (see Figure 2).
472 Developmentally, there was an increase in activation as a function of set-size in parietal cortex
473 such that 4-year-olds showed larger increases in activation over set-size relative to 3-year-olds.
474 Right frontal cortex showed an increase in activation between set-size 1 and 2. However, at set-
475 size 3, activation dropped back to set-size 1 levels. This suggests that activation within frontal
476 cortex increases across set-sizes that are within a child's ability but drops off as the number of
477 items exceeds capacity.

478
479 Tsujimoto, Yamamoto, Kawaguchi, Koizumi, and Sawaguchi (2004) used fNIRS to compare
480 frontal activation during a spatial WM task between a group of 5- and 6-year-olds and a group of
481 adults. Activation was observed in similar regions between age groups suggesting that children
482 engage similar neural mechanisms during spatial WM. However, EEG data suggests that young
483 children rely more on processing in posterior brain regions whereas adults rely more on anterior
484 brain regions (Barriga-Paulino, Rojas Benjumea, Rodríguez-Martínez, & Gómez González,
485 2015). Perlman, Huppert, and Luna (2016) used fNIRS to examine activation while 3- to 7-year-
486 old children performed a spatial WM task. Increases in activation were observed in bilateral
487 prefrontal cortex which corresponded to developmental improvements in spatial WM. Further,
488 activation in these regions increased as the delay during which the memory was maintained

increased. Taken together, these findings indicate that areas of frontal, parietal, and temporal cortex are involved in VWM throughout early childhood; however, children show increased activity in frontal areas during VWM tasks with increases in age, task demands, and task performance.

Insert Figure 3 here

Modeling VWM development

The development of VWM is marked by improvements in both the amount of information that can be maintained, and the precision of the representations being stored. Theories of VWM typically frame performance around the idea of a fixed number of slots measured as the capacity of VWM (Luck & Vogel, 1997; Pashler, 1988; Rouder, Morey, Morey, & Cowan, 2011; Zhang & Luck, 2008) or the allocation of resources to each representation indexed by the precision of representations in VWM (Bays, Catalao, & Husain, 2009; Bays & Husain, 2008; Wilken & Ma, 2004). More recent iterations of these theories have formulated Bayesian modeling methods to quantify and test predictions of capacity and precision models (e.g., Brady & Tenenbaum, 2013; Donkin et al., 2013; Kary et al., 2016; Rouder et al., 2008; Sims et al., 2012). Developmental improvements in VWM are thought to reflect increases in the capacity of VWM (Cowan, 2013; Riggs et al., 2006) or the precision of VWM representations (Heyes, Zokaei, van der Staaij, Bays, & Husain, 2012).

An alternative modeling approach focuses on the dynamics that give rise to the formation and maintenance of VWM representations. Improvements in performance described above have been formally explained using a neural process model. Dynamic neural field (DNF) architectures implement neural population dynamics that simulate the real-time dynamics of excitatory and

inhibitory interactions that underlie the formation and maintenance of visual representations. The neural fields are composed of neural units that are tuned to dimensions of visual information such as space, color, or shape. In these applications, developmental improvements in VWM performance are explained through increasing the strength of excitatory and inhibitory interactions. Starting with the A-not-B task, Thelen, Schöner, Scheier, and Smith, (2001) demonstrated how successful reaching on B trials can be supported through the ability to form an active representation of the B location that persists between the cueing and reaching events. In the context of the A-not-B task, stronger excitatory and inhibitory neural interactions allow the model to maintain an active representation of the B location after the cueing event and suppress the habit associated with reaching to the A location. This model has been extended to spatial recall tasks used in later development which require participants to maintain a spatial representation within a continuous space. Using increases in excitatory and inhibitory neural interactions as a proxy for development, this work explains increases in the precision of spatial VWM, drifts in VWM over increased delay times between encoding and reporting, changes in the shift toward midline to away from midline over development, and the influence of previous spatial memories on the dynamics of spatial VWM (Schutte & Spencer, 2009; Schutte et al., 2003; Simmering, Schutte, & Spencer, 2008).

Simmering (2016) describes a dynamic neural field (DNF) architecture that explains development on change detection tasks from early childhood through adulthood. In the model, active VWM representations are supported by local-excitatory/lateral-inhibitory neural interactions with populations of neural units that are tuned to metric perceptual dimensions such as color. The activation of VWM representations establishes an inhibitory perceptual filter on a

population of neural units that are responsible for encoding stimuli. Through these processes, the model can build active VWM representations that can be compared to subsequently presented stimuli. Specifically, stimuli that match items maintained in VWM will encounter strong inhibitory suppression, but stimuli that do not match items maintained in VWM will not encounter such inhibition. Based on the comparison of activation in VWM and visual stimuli presented during the comparison phase of the trial, the model generates active decisions corresponding to “same” and “different” responses. This model architecture simulated accuracy on same and different trials (i.e., hits, misses, false-alarms, and correct rejections) that matched the performance of children and adults. As neural interactions in the model were strengthened, model performance improved to match that of older age groups. Moreover, this model was able to address one of the primary factors that has limited our understanding of VWM development from infancy through early childhood and into adulthood. Specifically, the tasks used to probe VWM during these different age ranges rely on different types of responses. The use of preferential looking tasks during infancy and change detection tasks during childhood have led to inconsistencies in the estimated capacity between infancy, early childhood, and adulthood (see Figure 2). Simmering (2016) was able explain associations in performance between these tasks. Specifically, the same VWM system was coupled to a system that implemented looking dynamics and the model was allowed to freely look between inputs that matched the properties of the displays in the preferential looking task. Based on increases in the strength of neural interactions, the model matched performance in both tasks with a group of 3-, 4-, 5-year-olds and adults.

In summary, research in early childhood has uncovered various types of changes in VWM performance with spatial, color, and shape information. VWM processes in early childhood become more stable and precise. This is revealed by tasks that require recall of specific visual features and tasks that require comparison of remembered information with currently available stimuli. Neural process models suggest that the complex patterns of performance can emerge from changes in the strength of neural connectivity and demands imposed by different tasks.

Conclusion

VWM shows significant development across infancy and early childhood. Much of the developmental changes that occur across infancy in VSTM and VWM coincide with major gains in visual attention and perceptual processing. As the infant develops greater eye movement control in the first six months of postnatal development (Courage, Reynolds, & Richards, 2006; Johnson et al., 1991; Posner & Rothbart, 2013), their capacity for directing selective attention in an efficient and functional manner for supporting VWM processes increases dramatically. Similarly, as perceptual sensitivity increases significantly across this same age range, the input available to the infant for memory processing expands dramatically. However, VWM also influences visual search in infancy (Mitsven et al., 2018), and thus there are likely bidirectional effects between visual attention and VWM in infancy and beyond (Amso & Scerif, 2015; Astle & Scerif, 2011; Awh et al., 2006; Reynolds & Romano, 2016; Shimi et al., 2014). As can be seen in Figure 2, from 6 to 13 months of age, infants demonstrate a steady increase in VWM capacity (Ross-Sheehy et al., 2003, 2011). Within this age range, infants also show increased ability to bind objects and locations (Kaldy & Leslie, 2003, 2005). Gains in VWM capacity continue

through the early childhood years, paired with increased accuracy in VWM for spatial and featural information (Riggs et al., 2006; Simmering, 2012a, 2016; Simmering et al., 2015).

Although there is some disagreement in the underlying neural underpinnings of these developmental changes, the extant literature indicates areas of frontal, parietal, and temporal cortex as well as the dopaminergic neurotransmitter system are involved in VWM in infancy (Bacher et al., 2017; Baird et al., 2002; Bell, 2012; Wilcox et al., 2010, 2012, 2015) and childhood (Buss et al., 2014; Crone et al., 2006; Klingberg et al., 2002; Scherf, Sweeney, & Luna, 2006; Sweeney et al., 1996). Increases in frontal activity are found to occur with increases in age, task demands, and task performance (Buss et al., 2016; Perlman et al., 2016; Tsujimoto et al., 2004). Modeling work supports the likelihood that the strengthening of local-excitatory and lateral-inhibitory connections within this network is fundamentally involved in developmental gains in VWM (Schutte et al., 2003, 2008, 2009; Simmering, 2016; Thelen et al., 2001). Pressing issues for future research will be to specify how changes in performance across different measures of VWM during early childhood arise from the types of neural processes that have been uncovered, to specify the mechanisms that create changes in the neurocognitive system over development and how these mechanisms are influenced by experiences, and to understand how performance in these relatively simple VWM tasks are related to processing and performance in more complex aspects of cognition.

References

- Adams, R. J., & Courage, M. L. (1995). Development of chromatic discrimination in early infancy. *Behavioural Brain Research*, 67(1), 99–101. [https://doi.org/10.1016/0166-4328\(94\)00143-4](https://doi.org/10.1016/0166-4328(94)00143-4)
- Amso, D., & Scerif, G. (2015). The attentive brain: insights from developmental cognitive neuroscience. *Nature Reviews Neuroscience*, 16(10), 606–619. <https://doi.org/10.1038/nrn4025>
- Astle, D. E., & Scerif, G. (2011). Interactions between attention and visual short-term memory (VSTM): What can be learnt from individual and developmental differences? *Neuropsychologia*, 49(6), 1435–1445. <https://doi.org/10.1016/J.NEUROPSYCHOLOGIA.2010.12.001>
- Atkinson, J., Braddick, O., & Reed, N. E. J. (2012). The Developing Visual Brain : typical and atypical development Approximately half of your brain is used in everyday visual tasks Behavioural and electrophysiological measures each have, 1–27.
- Atkinson, R. C., & Shiffrin, R. M. (1968). Human Memory: A Proposed System and its Control Processes. In *Psychology of Learning and Motivation - Advances in Research and Theory* (Vol. 2, pp. 89–195). [https://doi.org/10.1016/S0079-7421\(08\)60422-3](https://doi.org/10.1016/S0079-7421(08)60422-3)
- Awh, E., Vogel, E. K., & Oh, S.-H. (2006). Interactions between attention and working memory. *Neuroscience*, 139(1), 201–208. <https://doi.org/10.1016/J.NEUROSCIENCE.2005.08.023>
- Bacher, L. F., Retz, S., Lindon, C., & Bell, M. A. (2017). Intraindividual and Interindividual Differences in Spontaneous Eye Blinking: Relationships to Working Memory Performance and Frontal EEG Asymmetry. *Infancy*, 22(2), 150–170. <https://doi.org/10.1111/infa.12164>
- Baddeley, A. D. (1986). *Working memory*. New York: Oxford University Press.

- 622 Baddeley, A. D. (2007). *Working Memory, Thought, and Action*. Oxford: Oxford University
623 Press.
- 624 Baddeley, A. D., & Hitch. (1974). Working memory. In *Psychology of Learning and Motivation*
625 (Vol. 8) (pp. 47–89). Academic Press. [https://doi.org/https://doi.org/10.1016/S0079-](https://doi.org/https://doi.org/10.1016/S0079-7421(08)60452-1)
626 7421(08)60452-1
- 627 Baddeley, A. D., & Hitch, G. J. (1994). Developments in the concept of working memory.
628 *Neuropsychology*, 8(4), 485–493. [https://doi.org/http://dx.doi.org/10.1037/0894-](https://doi.org/http://dx.doi.org/10.1037/0894-4105.8.4.485)
629 4105.8.4.485
- 630 Baddeley, A. D., & Logie, R. H. (1999). Working memory: The multiple-component model. In
631 *Models of working memory: Mechanisms of active maintenance and executive control. BT -*
632 *Models of working memory: Mechanisms of active maintenance and executive control.* (pp.
633 28–61). Cambridge University Press, New York, NY.
- 634 Bahrick, L. E., & Lickliter, R. (2014). Learning to Attend Selectively. *Current Directions in*
635 *Psychological Science*, 23(6), 414–420. <https://doi.org/10.1177/0963721414549187>
- 636 Baird, A. A., Kagan, J., Gaudette, T., Walz, K. A., Hershlag, N., & Boas, D. A. (2002). Frontal
637 Lobe Activation during Object Permanence: Data from Near-Infrared Spectroscopy.
638 *NeuroImage*, 16, 1120–1126. <https://doi.org/10.1006/nimg.2002.1170>
- 639 Banks, M. S., & Salapatek, P. (1981). Infant pattern vision: A new approach based on the
640 contrast sensitivity function. *Journal of Experimental Child Psychology*, 31(1), 1–45.
641 [https://doi.org/10.1016/0022-0965\(81\)90002-3](https://doi.org/10.1016/0022-0965(81)90002-3)
- 642 Barriga-Paulino, C. I., Rojas Benjumea, M. Á., Rodríguez-Martínez, E. I., & Gómez González,
643 C. M. (2015). Fronto–temporo–occipital activity changes with age during a visual working
644 memory developmental study in children, adolescents and adults. *Neuroscience Letters*,

- 645 599, 26–31. <https://doi.org/10.1016/J.NEULET.2015.05.017>
- 646 Bauer, P. J. (2009). The cognitive neuroscience of the development of memory. In *The*
647 *Development of Memory in Infancy and Childhood* (pp. 115–144). New York, NY:
648 Psychology Press.
- 649 Bays, P. M., Catalao, R. F. G., & Husain, M. (2009). The precision of visual working memory is
650 set by allocation of a shared resource. *Journal of Vision*, 9(10), 1–11.
651 <https://doi.org/10.1167/9.10.7.Introduction>
- 652 Bays, P. M., & Husain, M. (2008). Dynamic shifts of limited working memory resources in
653 human vision. *Science*, 321, 851–854. <https://doi.org/10.1126/science.1158023>
- 654 Bell, M. A. (2002). Power changes in infant EEG frequency bands during a spatial working
655 memory task. *Psychophysiology*, 39(4), S0048577201393174.
656 <https://doi.org/10.1017/S0048577201393174>
- 657 Bell, M. A. (2012). A Psychobiological Perspective on Working Memory Performance at 8
658 Months of Age. *Child Development*, 83(1), 251–265. [https://doi.org/10.1111/j.1467-](https://doi.org/10.1111/j.1467-8624.2011.01684.x)
659 [8624.2011.01684.x](https://doi.org/10.1111/j.1467-8624.2011.01684.x)
- 660 Bell, M. A., & Adams, S. E. (1999). Comparable performance on looking and reaching versions
661 of the A-not-B task at 8 months of age. *Infant Behavior and Development*, 22(2), 221–235.
662 [https://doi.org/10.1016/S0163-6383\(99\)00010-7](https://doi.org/10.1016/S0163-6383(99)00010-7)
- 663 Bell, M. A., & Fox, N. A. (1994). Brain development over the first year of life: Relations
664 between EEG frequency and coherence and cognitive and affective behaviors. In G.
665 Dawson & K. Fischer (Eds.), *Human Behavior and the Developing Brain* (pp. 314–345).
666 New York, NY: Guilford.
- 667 Bell, M. A., & Wolfe, C. D. (2007). Changes in brain functioning from infancy to early

childhood: evidence from EEG power and coherence working memory tasks.

Developmental Neuropsychology, 31(1), 21–38.

<https://doi.org/10.1080/87565640709336885>

Brady, T. F., & Tenenbaum, J. B. (2013). A probabilistic model of visual working memory:

Incorporating higher order regularities into working memory capacity estimates.

Psychological Review, 120(1), 85–109. <https://doi.org/10.1037/a0030779>

Brown, A. M., & Lindsey, D. T. (2009). Contrast insensitivity: The critical immaturity in infant

visual performance. *Optometry and Vision Science*, 86(6), 572–576.

<https://doi.org/10.1097/OPX.0b013e3181a72980>

Buss, A. T., Fox, N., Boas, D. A., & Spencer, J. P. (2014). Probing the early development of

visual working memory capacity with functional near-infrared spectroscopy. *NeuroImage*,

85 Pt 1, 314–25. <https://doi.org/10.1016/j.neuroimage.2013.05.034>

Colombo, J., & Fagen, J. W. (1990). *Individual differences in infancy: Reliability, stability,*

prediction. Individual differences in infancy: Reliability, stability, prediction. Lawrence

Erlbaum Associates, Inc, Hillsdale, NJ.

Courage, M. L., Reynolds, G. D., & Richards, J. E. (2006). Infants' Attention to Patterned

Stimuli: Developmental Change From 3 to 12 Months of Age. *Child Development*, 77(3),

680–695. <https://doi.org/10.1111/j.1467-8624.2006.00897.x>

Courtney, S. M., Ungerleider, L. G., Keil, K., & Haxby, J. V. (1997). Transient and sustained

activity in a distributed neural system for human working memory. *Nature*, 386(6625),

608–611. <https://doi.org/10.1038/386608a0>

Cowan, N. (1995). *Attention and memory: An integrated framework.* New York, NY: Oxford

University Press.

- 691 Cowan, N. (2005). *Working memory capacity. Essays in cognitive psychology*. Psychology Press,
692 New York, NY. <https://doi.org/http://dx.doi.org/10.4324/9780203342398>
- 693 Cowan, N. (2013). Short-term and working memory in childhood. In *The Wiley Handbook on the*
694 *Development of Children's Memory*. (pp. 202–229). West Sussex, UK: Wiley-Blackwell.
- 695 Crone, E. A., Wendelken, C., Donohue, S., van Leijenhorst, L., & Bunge, S. A. (2006).
696 Neurocognitive development of the ability to manipulate information in working memory.
697 *Proceedings of the National Academy of Sciences of the United States of America*, 103(24),
698 9315–20. <https://doi.org/10.1073/pnas.0510088103>
- 699 Cuevas, K., & Bell, M. A. (2010). Developmental progression of looking and reaching
700 performance on the A-not-B task. *Developmental Psychology*, 46(5), 1363–71.
701 <https://doi.org/10.1037/a0020185>
- 702 Cuevas, K., & Bell, M. A. (2011). EEG and ECG from 5 to 10 months of age: Developmental
703 changes in baseline activation and cognitive processing during a working memory task.
704 *International Journal of Psychophysiology*, 80(2), 119–128.
705 <https://doi.org/10.1016/J.IJPSYCHO.2011.02.009>
- 706 Diamond, A. (1985). Development of the Ability to Use Recall to Guide Action, as Indicated by
707 Infants' Performance on AB. *Child Development*, 56(4), 868.
708 <https://doi.org/10.2307/1130099>
- 709 Diamond, A. (1990). Rate of maturation of the hippocampus and the developmental progression
710 of children's performance on the delayed non-matching to sample and visual paired
711 comparison tasks. In A. Diamond (Ed.), *Development and Neural Bases of Higher*
712 *Cognitive Functions* (pp. 394–426). New York, NY: New York Academy of Sciences Press.
- 713 Dobson, V., & Teller, D. Y. (1978). Visual acuity in human infants: A review and comparison of

- 714 behavioral and electrophysiological studies. *Vision Research*, 18(11), 1469–1483.
715 [https://doi.org/10.1016/0042-6989\(78\)90001-9](https://doi.org/10.1016/0042-6989(78)90001-9)
- 716 Donkin, C., Nosofsky, R. M., Gold, J. M., & Shiffrin, R. M. (2013). Discrete-slots models of
717 visual working-memory response times. *Psychological Review*, 120(4), 873–902.
718 <https://doi.org/10.1037/a0034247>
- 719 Fantz, R. L. (1964). Visual Experience in Infants: Decreased Attention to Familiar Patterns
720 Relative to Novel Ones. *Science*. <https://doi.org/10.1126/science.146.3644.668>
- 721 Funahashi, S., Bruce, C. J., & Goldman-Rakic, P. S. (1989). Mnemonic coding of visual space in
722 the monkey's dorsolateral prefrontal cortex. *Journal of Neurophysiology*, 61(2), 331–49.
723 <https://doi.org/10.1152/jn.1989.61.2.331>
- 724 Fuster, J. M. (1997). *The Prefrontal Cortex: Anatomy, Physiology, and Neuropsychology of the*
725 *Frontal Lobes*. New York, NY: Raven Press.
- 726 Gilmore, R., & Johnson, M. H. (1995). Working Memory in Infancy: Six-Month-Olds'
727 Performance on Two Versions of the Oculomotor Delayed Response Task. *Journal of*
728 *Experimental Child Psychology*, 59(3), 397–418. <https://doi.org/10.1006/JECP.1995.1019>
- 729 Goldman-Rakic, P. (1995). Cellular basis of working memory. *Neuron*, 14(3), 477–485.
730 [https://doi.org/10.1016/0896-6273\(95\)90304-6](https://doi.org/10.1016/0896-6273(95)90304-6)
- 731 Hamer, R. D., Alexander, K. R., & Teller, D. Y. (1982). Rayleigh discriminations in young
732 human infants. *Vision Research*, 22(5), 575–7. [https://doi.org/10.1016/0042-](https://doi.org/10.1016/0042-6989(82)90116-X)
733 [6989\(82\)90116-X](https://doi.org/10.1016/0042-6989(82)90116-X)
- 734 Hassin, R. R., Bargh, J. A., Engell, A. D., & McCulloch, K. C. (2009). Implicit working
735 memory. *Consciousness and Cognition*, 18(3), 665–678.
736 <https://doi.org/10.1016/j.concog.2009.04.003>

- 737 Heyes, S., Zokaei, N., van der Staaij, I., Bays, P. M., & Husain, M. (2012). Development of
738 visual working memory precision in childhood. *Developmental Science*, 15(4), 528–539.
739 <https://doi.org/10.1111/j.1467-7687.2012.01148.x>
- 740 Hofstadter, M., & Reznick, J. S. (1996). Response Modality Affects Human Infant Delayed-
741 Response Performance. *Child Development*, 67(2), 646. <https://doi.org/10.2307/1131838>
- 742 Johnson, M. H. (1990). Cortical maturation and the development of visual attention in early
743 infancy. *Journal of Cognitive Neuroscience*, 2(2), 81–95.
744 <https://doi.org/10.1162/jocn.1990.2.2.81>
- 745 Johnson, M. H. (1992). Imprinting and the Development of Face Recognition: From Chick to
746 Man. *Current Directions in Psychological Science*, 1(2), 52–55.
747 <https://doi.org/10.1111/1467-8721.ep11509740>
- 748 Johnson, M. H. (2011). *Developmental Cognitive Neuroscience* (3rd Editio). West Sussex, UK:
749 Blackwell Publishing.
- 750 Johnson, M. H., Posner, M. I., & Rothbart, M. K. (1991). Components of visual orienting in
751 early infancy: Contingency learning, anticipatory looking, and disengaging. *Journal of*
752 *Cognitive Neuroscience*, 3(4), 335–344.
753 <https://doi.org/http://dx.doi.org/10.1162/jocn.1991.3.4.335>
- 754 Joyce, A. W. (2016). Implicit Working Memory: Implications for Assessment and Treatment.
755 *Applied Neuropsychology: Child*, 5(3), 223–234.
756 <https://doi.org/10.1080/21622965.2016.1167497>
- 757 Kaldy, Z., Guillory, S. B., & Blaser, E. (2016). Delayed Match Retrieval: a novel anticipation-
758 based visual working memory paradigm. *Developmental Science*, 19(6), 892–900.
759 <https://doi.org/10.1111/desc.12335>

- 760 Kaldy, Z., & Leslie, A. M. (2003). Identification of objects in 9-month-old infants: integrating
761 “what” and “where” information. *Developmental Science*, 6(3), 360–373.
762 <https://doi.org/10.1111/1467-7687.00290>
- 763 Kaldy, Z., & Leslie, A. M. (2005). A memory span of one? Object identification in 6.5-month-
764 old infants. *Cognition*, 97(2), 153–177. <https://doi.org/10.1016/J.COGNITION.2004.09.009>
- 765 Kaldy, Z., & Sigala, N. (2004). The neural mechanisms of object working memory: what is
766 where in the infant brain? *Neuroscience & Biobehavioral Reviews*, 28(2), 113–121.
767 <https://doi.org/10.1016/J.NEUBIOREV.2004.01.002>
- 768 Kary, A., Taylor, R., & Donkin, C. (2016). Using Bayes factors to test the predictions of models:
769 A case study in visual working memory. *Journal of Mathematical Psychology*, 72, 210–219.
770 <https://doi.org/10.1016/j.jmp.2015.07.002>
- 771 Kibbe, M. M. (2015). Varieties of visual working memory representation in infancy and beyond.
772 *Current Directions in Psychological Science*, 24(6), 433–439.
773 <https://doi.org/http://dx.doi.org/10.1177/0963721415605831>
- 774 Klingberg, T., Forssberg, H., & Westerberg, H. (2002). Increased brain activity in frontal and
775 parietal cortex underlies the development of visuospatial working memory capacity during
776 childhood. *Journal of Cognitive Neuroscience*, 14(1), 1–10.
777 <https://doi.org/10.1162/089892902317205276>
- 778 Kwon, M.-K., Luck, S. J., & Oakes, L. M. (2014). Visual Short-Term Memory for Complex
779 Objects in 6- and 8-Month-Old Infants. *Child Development*, 85(2), 564–577.
780 <https://doi.org/10.1111/cdev.12161>
- 781 Luck, S. J., & Vogel, E. K. (1997). The capacity of visual working memory for features and
782 conjunctions. *Nature*, 390, 279–281. <https://doi.org/10.1038/36846>

- 783 Luck, S. J., & Vogel, E. K. (2013). Visual working memory capacity: from psychophysics and
784 neurobiology to individual differences. *Trends in Cognitive Sciences*, 17(8), 391–400.
785 <https://doi.org/10.1016/j.tics.2013.06.006>
- 786 Mitsven, S. G., Cantrell, L. M., Luck, S. J., & Oakes, L. M. (2018). Visual short-term memory
787 guides infants' visual attention. *Cognition*, 177, 189–197.
788 <https://doi.org/10.1016/J.COGNITION.2018.04.016>
- 789 Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, a H., Howerter, A., & Wager, T. D.
790 (2000). The unity and diversity of executive functions and their contributions to complex
791 “Frontal Lobe” tasks: a latent variable analysis. *Cognitive Psychology*, 41(1), 49–100.
792 <https://doi.org/10.1006/COGP.1999.0734>
- 793 Moher, M., & Feigenson, L. (2013). Factors influencing infants' ability to update object
794 representations in memory. *Cognitive Development*, 28(3), 272–289.
795 <https://doi.org/10.1016/j.cogdev.2013.04.002>
- 796 Nagata, Y., & Dannemiller, J. L. (1996). The selectivity of motion-driven visual attention in
797 infants. *Child Development*, 67(6), 2608–2620. [https://doi.org/10.1111/j.1467-](https://doi.org/10.1111/j.1467-8624.1996.tb01878.x)
798 [8624.1996.tb01878.x](https://doi.org/10.1111/j.1467-8624.1996.tb01878.x)
- 799 Nelson, C. A. (1995). The Ontogeny of Human Memory: A Cognitive Neuroscience Perspective.
800 *Developmental Psychology*, 31(5), 723–738. <https://doi.org/10.1037/0012-1649.31.5.723>
- 801 Oakes, L. M., Baumgartner, H. A., Barrett, F. S., Messenger, I. M., & Luck, S. J. (2013).
802 Developmental changes in visual short-term memory in infancy: Evidence from eye-
803 tracking. *Frontiers in Psychology*, 4(OCT), 1–13. <https://doi.org/10.3389/fpsyg.2013.00697>
- 804 Oakes, L. M., Messenger, I. M., Ross-Sheehy, S., & Luck, S. J. (2009). New evidence for rapid
805 development of colour-location binding in infants' visual short-term memory. *Visual*

- 806 *Cognition*, 17(1–2), 67–82. <https://doi.org/10.1080/13506280802151480>
- 807 Oakes, L. M., Ross-Sheehy, S., & Luck, S. J. (2006). Rapid development of feature binding in
808 visual short-term memory. *Psychological Science*, 17(9), 781–787.
809 <https://doi.org/10.1111/j.1467-9280.2006.01782.x>
- 810 Pashler, H. (1988). Familiarity and the detection of change in visual displays. *Perception and*
811 *Psychophysics*, 44, 369–378.
- 812 Pelphrey, K. A., & Reznick, J. S. (2003). Working memory in infancy. In R. V. Kail (Ed.),
813 *Advances in Child Behavior*, Vol. 31 (pp. 173–227). San Diego, CA: Academic Press.
- 814 Pelphrey, K. A., Reznick, J. S., Davis Goldman, B., Sasson, N., Morrow, J., Donahoe, A., &
815 Hodgson, K. (2004). Development of visuospatial short-term memory in the second half of
816 the 1st year. *Developmental Psychology*, 40(5), 836–51. [https://doi.org/10.1037/0012-](https://doi.org/10.1037/0012-1649.40.5.836)
817 1649.40.5.836
- 818 Perlman, S. B., Huppert, T. J., & Luna, B. (2016). Functional Near-Infrared Spectroscopy
819 Evidence for Development of Prefrontal Engagement in Working Memory in Early
820 Through Middle Childhood. *Cerebral Cortex*, 26(6), 2790–2799.
821 <https://doi.org/10.1093/cercor/bhv139>
- 822 Posner, M. I., & Rothbart, M. K. (2013). Development of attention networks. In B. R. Kar (Ed.),
823 *Cognition and Brain Development: Converging Evidence from Various Methodologies* (pp.
824 61–83). Washington, DC: American Psychological Association.
- 825 Raghubar, K. P., Barnes, M. A., & Hecht, S. A. (2010). Working memory and mathematics: A
826 review of developmental, individual difference, and cognitive approaches. *Learning and*
827 *Individual Differences*, 20(2), 110–122. <https://doi.org/10.1016/J.LINDIF.2009.10.005>
- 828 Rao, S. G., Williams, G. V., & Goldman-Rakic, P. S. (1999). Isodirectional Tuning of Adjacent

- 829 Interneurons and Pyramidal Cells During Working Memory: Evidence for Microcolumnar
830 Organization in PFC. *Journal of Neurophysiology*, 81(4), 1903–1916.
831 <https://doi.org/10.1152/jn.1999.81.4.1903>
- 832 Reynolds, G. D. (2015). Infant visual attention and object recognition. *Behavioural Brain*
833 *Research*, 285, 34–43. <https://doi.org/10.1016/J.BBR.2015.01.015>
- 834 Reynolds, G. D., & Richards, J. E. (2005). Familiarization, attention, and recognition memory in
835 infancy: an event-related potential and cortical source localization study. *Developmental*
836 *Psychology*, 41(4), 598–615. <https://doi.org/10.1037/0012-1649.41.4.598>
- 837 Reynolds, G. D., & Richards, J. E. (2017). Infant Visual Attention and Stimulus Repetition
838 Effects on Object Recognition. *Child Development*. <https://doi.org/10.1111/cdev.12982>
- 839 Reynolds, G. D., & Romano, A. C. (2016). The Development of Attention Systems and Working
840 Memory in Infancy. *Frontiers in Systems Neuroscience*, 10, 15.
841 <https://doi.org/10.3389/fnsys.2016.00015>
- 842 Reznick, J. S., Morrow, J. D., Goldman, B. D., & Snyder, J. (2004). The Onset of Working
843 Memory in Infants. *Infancy*, 6(1), 145–154. https://doi.org/10.1207/s15327078in0601_7
- 844 Richards, J. E. (1997). Effects of attention on infants' preference for briefly exposed visual
845 stimuli in the paired-comparison recognition-memory paradigm. *Developmental*
846 *Psychology*, 33(1), 22–31. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/9050387>
- 847 Riggs, K. J., McTaggart, J., Simpson, A., & Freeman, R. P. J. (2006). Changes in the capacity of
848 visual working memory in 5- to 10-year-olds. *Journal of Experimental Child Psychology*,
849 95(1), 18–26. <https://doi.org/10.1016/J.JECP.2006.03.009>
- 850 Roder, B. J., Bushnell, E. W., & Sasseville, A. M. (2000). Infants' Preferences for Familiarity
851 and Novelty during the Course of Visual Processing. *Infancy*, 1(4), 491–507.

https://doi.org/10.1207/S15327078IN0104_9

Rose, S. A., Feldman, J. F., & Jankowski, J. J. (2004). Infant visual recognition memory.

Developmental Review, 24(1), 74–100. <https://doi.org/10.1016/J.DR.2003.09.004>

Rose, S. A., Feldman, J. F., Jankowski, J. J., & Rossem, R. Van. (2008). A cognitive cascade in infancy: Pathways from prematurity to later mental development. *Intelligence*, 36(4), 367–378. <https://doi.org/10.1016/J.INTELL.2007.07.003>

Ross-Sheehy, S., Oakes, L. M., & Luck, S. J. (2003). The development of visual short-term memory capacity in infants. *Child Development*, 74, 1807–1822.

Ross-Sheehy, S., Oakes, L. M., & Luck, S. J. (2011). Exogenous attention influences visual short-term memory in infants. *Dev Sci*, 14(3), 490–501.

Rouder, J. N., Morey, R. D., Cowan, N., Zwilling, C. E., Morey, C. C., & Pratte, M. S. (2008). An assessment of fixed-capacity models of visual working memory. *Proceedings of the National Academy of Sciences of the United States of America*, 105(16), 5975–9. <https://doi.org/10.1073/pnas.0711295105>

Rouder, J. N., Morey, R. D., Morey, C. C., & Cowan, N. (2011). How to measure working memory capacity in the change detection paradigm. *Psychonomic Bulletin & Review*, 18(2), 324–330. <https://doi.org/10.3758/s13423-011-0055-3>

Rovee-Collier, C., & Cuevas, K. (2009). Multiple memory systems are unnecessary to account for infant memory development: An ecological model. *Developmental Psychology*, 45(1), 160–174. <https://doi.org/10.1037/a0014538>

Scherf, K. S., Sweeney, J. A., & Luna, B. (2006). Brain Basis of Developmental Change in Visuospatial Working Memory. *Journal of Cognitive Neuroscience*, 18(7), 1045–1058. <https://doi.org/10.1162/jocn.2006.18.7.1045>

- 875 Scherf, K. S., Sweeney, J. A., & Luna, B. (2006). Brain basis of developmental change in
876 visuospatial working memory. *Journal of Cognitive Neuroscience*, 18(7), 1045–58.
877 <https://doi.org/10.1162/jocn.2006.18.7.1045>
- 878 Schutte, A. R., & Spencer, J. P. (2009). Tests of the dynamic field theory and the spatial
879 precision hypothesis: capturing a qualitative developmental transition in spatial working
880 memory. *Journal of Experimental Psychology. Human Perception and Performance*, 35(6),
881 1698–725. <https://doi.org/10.1037/a0015794>
- 882 Schutte, A. R., Spencer, J. P., & Schoner, G. (2003). Testing the Dynamic Field Theory:
883 Working Memory for Locations Becomes More Spatially Precise Over Development. *Child*
884 *Development*, 74(5), 1393–1417.
- 885 Shimi, A., Nobre, A. C., Astle, D., & Scerif, G. (2014). Orienting Attention Within Visual Short-
886 Term Memory: Development and Mechanisms. *Child Development*, 85(2), 578–592.
887 <https://doi.org/10.1111/cdev.12150>
- 888 Simmering, V. R. (2012). The development of visual working memory capacity during early
889 childhood. *Journal of Experimental Child Psychology*, 111(4), 695–707.
890 <https://doi.org/10.1016/J.JECP.2011.10.007>
- 891 Simmering, V. R. (2016). Working memory in context: Modeling dynamic processes of
892 behavior, memory, and development. *Monographs of the Society for Research in Child*
893 *Development*, 81(3), 1–166. <https://doi.org/10.1111/mono.12249>
- 894 Simmering, V. R., & Miller, H. E. (2016). Developmental improvements in the resolution and
895 capacity of visual working memory share a common source. *Attention, Perception, &*
896 *Psychophysics*, 78(6), 1538–1555. <https://doi.org/10.3758/s13414-016-1163-y>
- 897 Simmering, V. R., Miller, H. E., & Bohache, K. (2015). Different developmental trajectories

across feature types support a dynamic field model of visual working memory development.

Attention, Perception, & Psychophysics, 77(4), 1170–1188. [https://doi.org/10.3758/s13414-](https://doi.org/10.3758/s13414-015-0832-6)

[015-0832-6](https://doi.org/10.3758/s13414-015-0832-6)

Simmering, V. R., Schutte, A. R., & Spencer, J. P. (2008). Generalizing the dynamic field theory of spatial cognition across real and developmental time scales. *Brain Research*, 1202, 68–86.

Sims, C. R., Jacobs, R. A., & Knill, D. C. (2012). An ideal observer analysis of visual working memory. *Psychological Review*, 119(4), 807–830. <https://doi.org/10.1037/a0029856>

Smith, L. B., Thelen, E., Titzer, R., & McLin, D. (1999). Knowing in the context of acting : The task dynamics of the A-not-B error. *Psychological Review*, 106(2), 235–260.

Spencer, J. P., & Schutte, A. R. (2004). Unifying Representations and Responses: Perseverative biases arise from a single behavioral system. *Psychological Science*, 15(3), 187–193.

Sreenivasan, K. K., Curtis, C. E., & D’Esposito, M. (2014). Revisiting the role of persistent neural activity during working memory. *Trends in Cognitive Sciences*, 18(2), 82–89.

<https://doi.org/10.1016/J.TICS.2013.12.001>

Sweeney, J. A., Mintun, M. A., Kwee, S., Wiseman, M. B., Brown, D. L., Rosenberg, D. R., & Carl, J. R. (1996). Positron emission tomography study of voluntary saccadic eye

movements and spatial working memory. *Journal of Neurophysiology*, 75(1), 454–68.

<https://doi.org/10.1152/jn.1996.75.1.454>

Thelen, E., Schöner, G., Scheier, C., & Smith, L. B. (2001). The dynamics of embodiment: a field theory of infant perseverative reaching. *The Behavioral and Brain Sciences*, 24(1), 1-34-86. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/11515285>

Tsujimoto, S., Kuwajima, M., & Sawaguchi, T. (2007). Developmental fractionation of working

- 921 memory and response inhibition during childhood. *Experimental Psychology*, 54(1), 30–7.
922 <https://doi.org/10.1027/1618-3169.54.1.30>
- 923 Tsujimoto, S., Yamamoto, T., Kawaguchi, H., Koizumi, H., & Sawaguchi, T. (2004). Prefrontal
924 cortical activation associated with working memory in adults and preschool children: An
925 event-related optical topography study. *Cerebral Cortex*, 14(7), 703–712.
- 926 Vogel, E. K., McCollough, A. W., & Machizawa, M. G. (2005). Neural measures reveal
927 individual differences in controlling access to working memory. *Nature*, 438(7067), 500–
928 503. <https://doi.org/10.1038/nature04171>
- 929 Wass, S. V., Scerif, G., & Johnson, M. H. (2012). Training attentional control and working
930 memory - Is younger, better? *Developmental Review*, 32(4), 360–387.
931 <https://doi.org/10.1016/j.dr.2012.07.001>
- 932 Watanabe, M., Kodama, T., & Hikosaka, K. (1997). Increase of Extracellular Dopamine in
933 Primate Prefrontal Cortex During a Working Memory Task. *Journal of Neurophysiology*,
934 78(5), 2795–2798. <https://doi.org/10.1152/jn.1997.78.5.2795>
- 935 Wiebe, S. A., Sheffield, T., Nelson, J. M., Clark, C. a C., Chevalier, N., & Espy, K. A. (2011).
936 The structure of executive function in 3-year-olds. *Journal of Experimental Child*
937 *Psychology*, 108(3), 436–52.
- 938 Wijeakumar, S., Huppert, T. J., Magnotta, V. A., Buss, A. T., & Spencer, J. P. (2017). Validating
939 an image-based fNIRS approach with fMRI and a working memory task. *NeuroImage*, 147.
940 <https://doi.org/10.1016/j.neuroimage.2016.12.007>
- 941 Wilcox, T., & Biondi, M. (2015). fNIRS in the developmental sciences. *Wiley Interdisciplinary*
942 *Reviews: Cognitive Science*, 6(3), 263–283. <https://doi.org/10.1002/wcs.1343>
- 943 Wilcox, T., Haslup, J. A., & Boas, D. A. (2010). Dissociation of processing of featural and

- 944 spatiotemporal information in the infant cortex. *NeuroImage*, 53(4), 1256–1263.
945 <https://doi.org/10.1016/J.NEUROIMAGE.2010.06.064>
- 946 Wilcox, T., Stubbs, J., Hirshkowitz, A., & Boas, D. A. (2012). Functional activation of the infant
947 cortex during object processing. *NeuroImage*, 62(3), 1833–1840.
948 <https://doi.org/10.1016/J.NEUROIMAGE.2012.05.039>
- 949 Wilken, P., & Ma, W. J. (2004). A detection theory account of change detection. *Journal of*
950 *Vision*, 4, 1120–1135.
- 951 Yamaguchi, M., & Feigenson, L. (2008). Selective attention of working memory contents by 11-
952 month-old infants. *Visual Cognition*, 16(8), 1144–1147.
- 953 Yantis, S., & Jonides, J. (1990). Abrupt visual onsets and selective attention: voluntary versus
954 automatic allocation. *Journal of Experimental Psychology. Human Perception and*
955 *Performance*, 16(1), 121–134. <https://doi.org/10.1037/0096-1523.16.1.121>
- 956 Zhang, W., & Luck, S. J. (2008). Discrete fixed-resolution representations in visual working
957 memory. *Nature*, 453, 233–235.
- 958
- 959

Figure Legends

Figure 1. Schematic of a set size 3 trial (top panel) and preference for the changing stream by age and set size (bottom panel). Note: Error bars = 1 SEM. Significant difference from chance (.5) denoted as **= $p < .01$, ***= $p < .001$. Figure adapted from Ross-Sheehy, Oakes and Luck, 2003.

Figure 2. General summary of developmental gains in VWM performance by task and age as discussed in the literature review. The bars represent the age range each task has been typically used with, and the dashed vertical line indicates the transition from the use of infant measures to child measures between 1 to 3 years of age. The darkness of the bar at each age point shown on the x-axis is relative to peak performance on the task the bar represents. The darkest point in each bar represents the approximate age participants reach peak performance on that task. The numbers to the left and right of the bars representing capacity measures indicate the average set size participants demonstrate evidence of detecting changes at the earliest and oldest ages tested.

Figure 3. Concentrations of OxyHb (dark lines) and Deoxy-Hb (light colors) for channels showing an effect of set-size (SS). The inset image shows the locations of the activated channels on frontal (left plot), left parietal (bottom three plots), and right parietal (right plot) regions. The shaded regions are the expected peak response to the memory array (blue shading) and test array (pink shading). Figure adapted from Buss et al., 2014.





